

Predicting COVID-19 Illness Severity: Identifying Key Risk Factors Using Machine Learning Classification Methods

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Introduction

This report analyzes data from a COVID-19 cohort study collecting demographic and clinical information among participants infected with COVID-19. The analysis has three objectives: (1) to compare the predictive performance of complex machine learning models (AdaBoost and Support Vector Machine) against simpler models (logistic regression and decision tree), (2) to develop an individual risk score for severe COVID-19 infection based on predicted probabilities from the best-performing model, and (3) to identify the key demographic and clinical factors that predict severe infection and characterize how they influence risk.

Methods

A random sample of 1,000 participants was drawn from the full dataset of 10,000 using a pre-specified random seed. The sample was split into a training set ($n = 800$, 80%) and a test set ($n = 200$, 20%) for model development and evaluation. The outcome variable of interest is COVID-19 illness severity, coded as a binary variable (severe vs. not severe). Of the 1,000 sampled participants, 321 (32.1%) were classified as severe and 679 (67.9%) as not severe, indicating a moderately imbalanced dataset.

Prior to model training, exploratory data analysis was performed on the sampled dataset. For continuous predictors – age, body mass index (BMI), systolic blood pressure (SBP), LDL cholesterol (LDL), and depression score – means and standard deviations were calculated. Percentages were computed for categorical variables including gender (1 = Male, 0 = Female), vaccination status (0 = unvaccinated, 1 = vaccinated), hypertension (0 = No, 1 = Yes), diabetes (0 = No, 1 = Yes), smoking status (0 = never, 1 = former, 2 = current), and race (1- 4). Summary statistics are presented in Table 1. Distributions of continuous and categorical predictors stratified by severity group are visualized in Figure 1.

Four classification models were trained on the training set using 10-fold cross-validation with AUC as the performance metric. AdaBoost and SVM with a radial kernel were selected as the complex models, and their predictive performance was compared against simpler baseline models (logistic regression and decision tree) to assess whether the added complexity yields meaningful improvements. All models were trained using the caret package in R. Logistic regression requires no hyperparameter tuning and was included as a linear baseline. The decision tree was tuned over the complexity parameter (cp), which controls the degree of pruning, across a grid of 100 values on a log scale. AdaBoost was tuned over four parameters: number of boosting iterations (n.trees), tree depth (interaction.depth), learning rate (shrinkage), and minimum observations per node (n.minobsinnode). The SVM with radial kernel was tuned over the cost parameter (C), which controls the margin width, and the kernel width parameter (sigma). Cross-validation tuning plots for all models are shown in figure 2.

Cross-validated performance was compared across all four models using the resamples() function in caret. The model with the highest mean cross-validated AUC was selected as the final model. All four models were then evaluated on the held-out test set, with performance reported via ROC curves and AUC.

The best-performing model was then subjected to further interpretation. Variable importance was assessed using relative influence, which measures the contribution of each predictor to the model's predictions based on the reduction in loss function attributable to each variable across all boosting iterations. Partial dependence plots (PDPs) were constructed for the top three predictors to visualize how each variable influences the predicted probability of severe infection, while averaging over all other predictors. Finally, the predicted probability of severe infection from the model was used as an individual risk score, with higher values indicating greater predicted risk of severe illness.

Results

Exploratory Data Analysis

Table 1 presents summary statistics stratified by severity group. Severe cases tended to be older (mean age 61.0 vs. 59.5 years), had higher mean SBP (132.6 vs. 128.5 mmHg), and higher mean

BMI (28.2 vs. 27.4 kg/m²). The most significant difference was in vaccination status – only 20.9% of severe cases were vaccinated compared to 80.4% of not severe cases. Severe cases also had a higher prevalence of hypertension (59.8% vs. 38.7%). LDL, depression score, gender, diabetes, race, and smoking status were largely similar between groups (Figure 1).

Model Training & Selection

Table 2 and Figure 3 summarize the cross-validated performance of all four models. AdaBoost achieved the highest mean cross-validated AUC (0.868), followed by logistic regression (0.861), SVM-Radial (0.859), and decision tree (0.833). Based on these results, AdaBoost was selected as the final model. The CV performance of the complex models (AdaBoost and SVM-Radial) was modestly higher than the simpler models, though the differences were small.

All four models performed well on the held-out test set (Figure 4). AdaBoost and SVM-Radial achieved the highest test AUC (0.918 each), followed by logistic regression (0.911) and decision tree (0.896). These results are consistent with the cross-validation findings, confirming that the complex models modestly outperform the simpler baselines. However, the differences in AUC were small across all models, suggesting that most of the predictive signal in this dataset can be captured by a simple linear model. Notably, logistic regression performed comparably to the complex models, with only a 0.007 difference in AUC from AdaBoost in both cross-validation and test set performance.

Risk Scores

The predicted probabilities from the AdaBoost model were used as individual risk scores for severe COVID-19 infection (Tables 3 and 4). Among the top 10 highest-risk patients in the test set, all were correctly identified as truly severe, with predicted probabilities ranging from 0.813 to 0.904 (Table 3). The risk scores showed clear separation between severity groups: severe patients had a mean predicted probability of 0.602, compared to 0.184 for not severe patients (Table 4), demonstrating that the AdaBoost model produces a clinically meaningful risk score.

Key Predictors of Severity

Figure 5 shows variable importance from the AdaBoost model. Vaccination status was by far the most influential predictor, accounting for 54.98% of relative influence, followed by SBP (21.39%) and BMI (12.58%). Age contributed modestly (5.19%), while all remaining predictors (depression, LDL, gender, race, smoking, hypertension, and diabetes) contributed less than 4% combined.

Partial dependence plots for the top three predictors are shown in Figure 6. Unvaccinated patients had a predicted probability of severe infection of approximately 0.59, compared to 0.15 for vaccinated patients, highlighting the strong protective effect of vaccination. For SBP, predicted severity risk remained low below approximately 125 mmHg and increased sharply between 125–140 mmHg before plateauing around 0.45, suggesting a threshold effect. A similar pattern was observed for BMI, with risk increasing markedly above approximately 25 kg/m².

Discussion

Based on cross-validated AUC, AdaBoost was selected as the best performing model, modestly outperforming logistic regression, SVM-Radial, and decision tree. However, the small differences in CV AUC suggest that the relationship between predictors and COVID-19 severity is largely linear and additive, and that a simple logistic regression model may be sufficient where interpretability is valued. The predicted probabilities from AdaBoost served as a clinically meaningful risk score, with severe patients having a mean predicted probability of 0.602 compared to 0.184 for not severe patients, and all top 10 highest-risk patients correctly identified as truly severe. Vaccination status was the dominant predictor of severity, with unvaccinated patients having approximately four times higher predicted risk than vaccinated patients. SBP and BMI were the next most important predictors, both exhibiting nonlinear threshold effects.

Appendix

Table 1. Summary Statistics by severity group

	Severe	Not Severe
severity	severe	not_severe
n	321	679
age_mean_sd	61 (4.2)	59.5 (4.4)
bmi_mean_sd	28.2 (2.8)	27.4 (2.6)
sbp_mean_sd	132.6 (7.3)	128.5 (8)
ldl_mean_sd	111.2 (18.7)	109.3 (20.1)
dep_mean_sd	7 (1.8)	6.9 (2)
pct_male	49.8%	49.6%
pct_vaccinated	20.9%	80.4%
pct_htn	59.8%	38.7%
pct_diabetes	13.1%	15.2%

Figure 1. Distributions of predictors stratified by COVID-19 severity. (a) Boxplots of continuous predictors: age, BMI, SBP, LDL, and depression score. (b) Stacked bar charts of categorical predictors: diabetes, gender, hypertension, race, smoking status, and vaccination status.

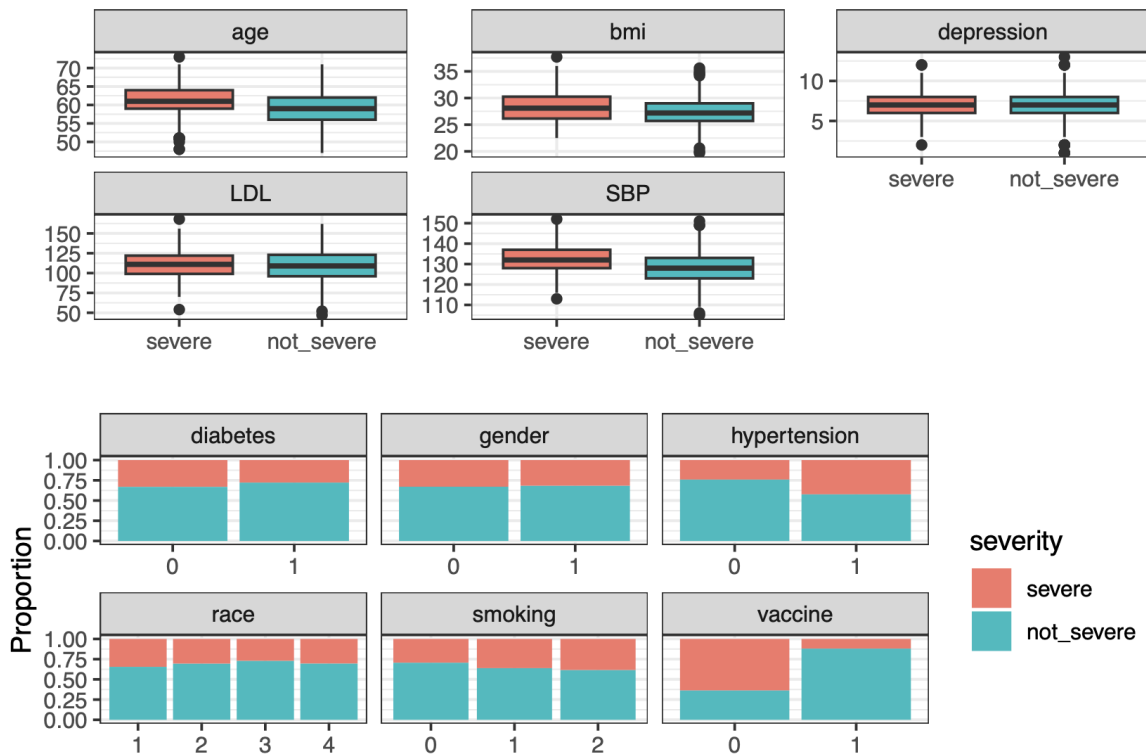


Figure 2. Cross-validation tuning plots for (a) decision tree: ROC as a function of the complexity parameter (cp) on a log scale, (b) AdaBoost: ROC as a function of number of boosting iterations across shrinkage values and tree depths, and (c) SVM-Radial: ROC as a function of cost (C) across sigma values.

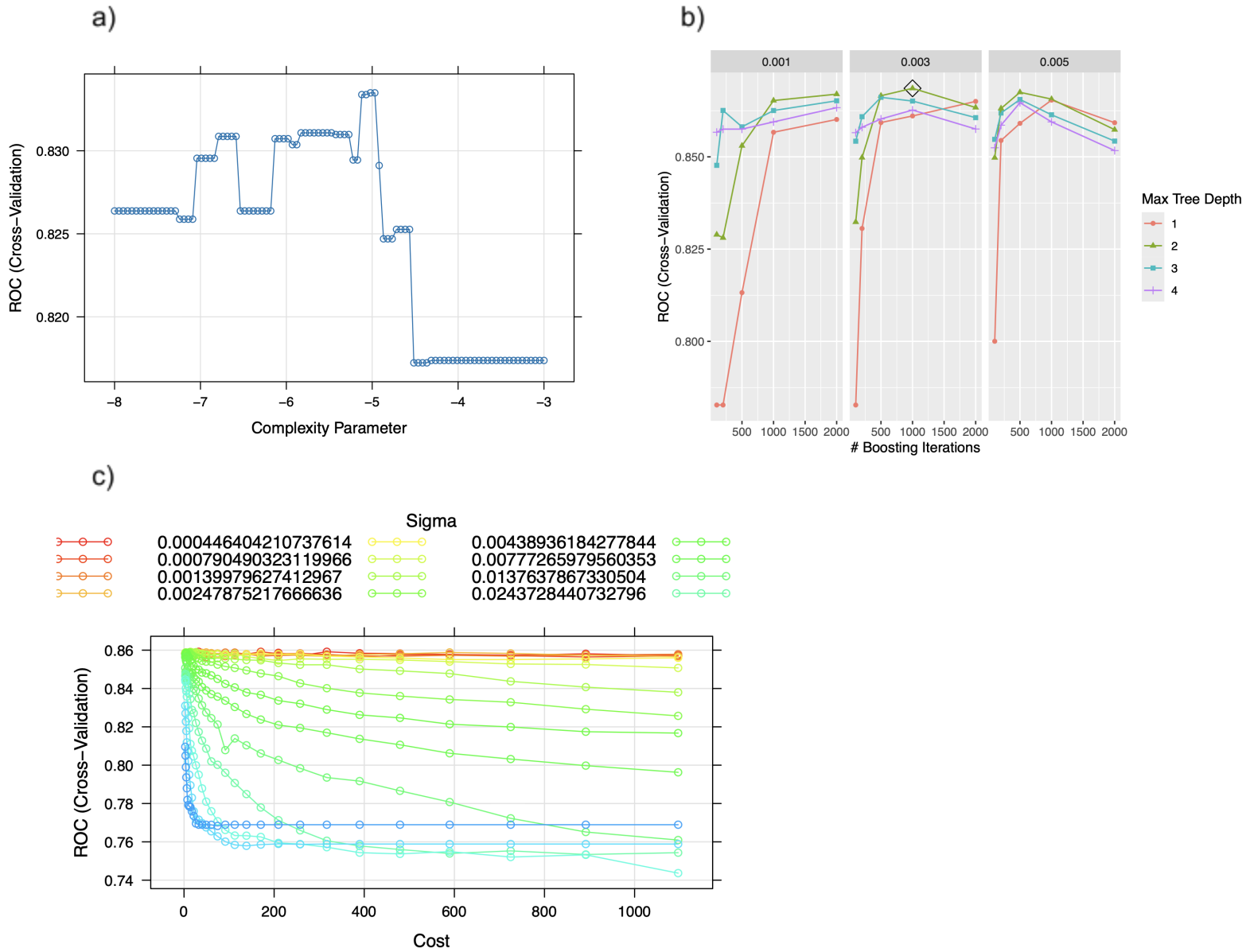


Figure 3. Boxplot of cross-validated AUC across 10 folds for all four models.

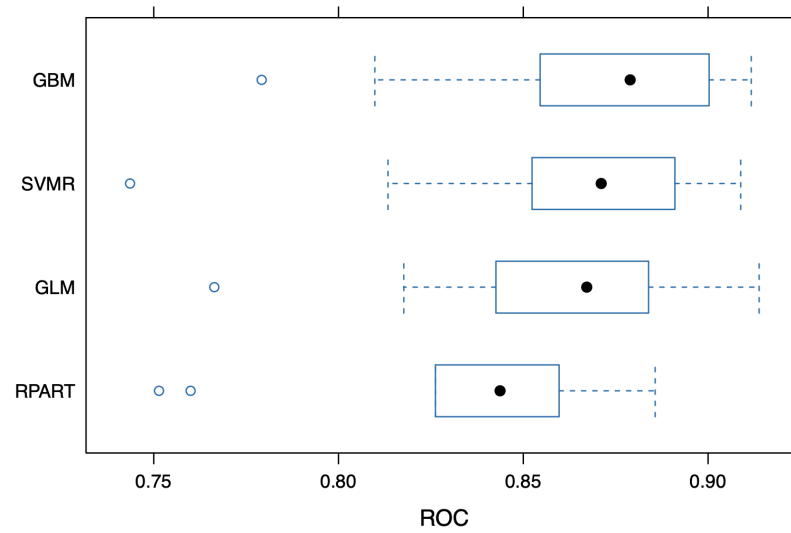


Table 2. Cross-validated model performance (AUC, sensitivity, specificity) across 10 folds for all four classification models.

Model	Mean AUC	Median AUC	Mean Sensitivity	Mean Specificity
Logistic Regression	0.861	0.867	0.676	0.885
Decision Tree	0.833	0.844	0.679	0.893
AdaBoost	0.868	0.861	0.660	0.915
SVM-Radial	0.859	0.871	0.734	0.810

Figure 4. ROC curves on the test set for all four models with corresponding AUC values.

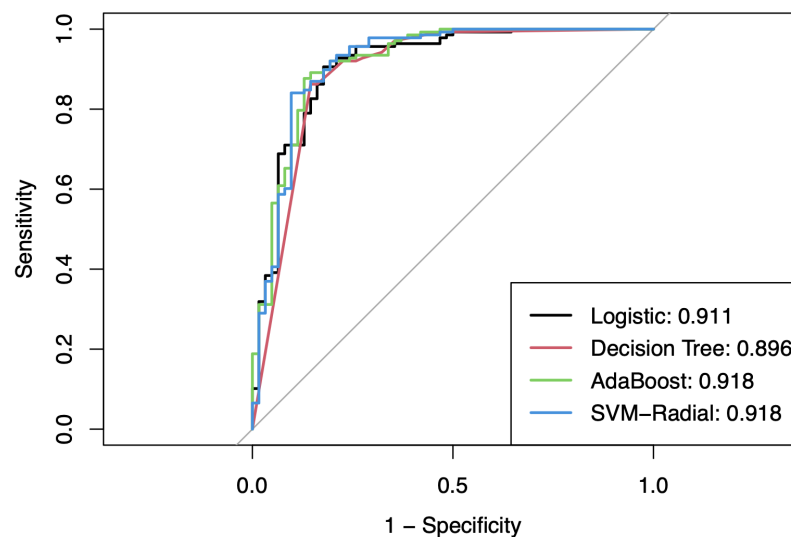


Figure 5. Variable importance from the AdaBoost model measured by relative influence.

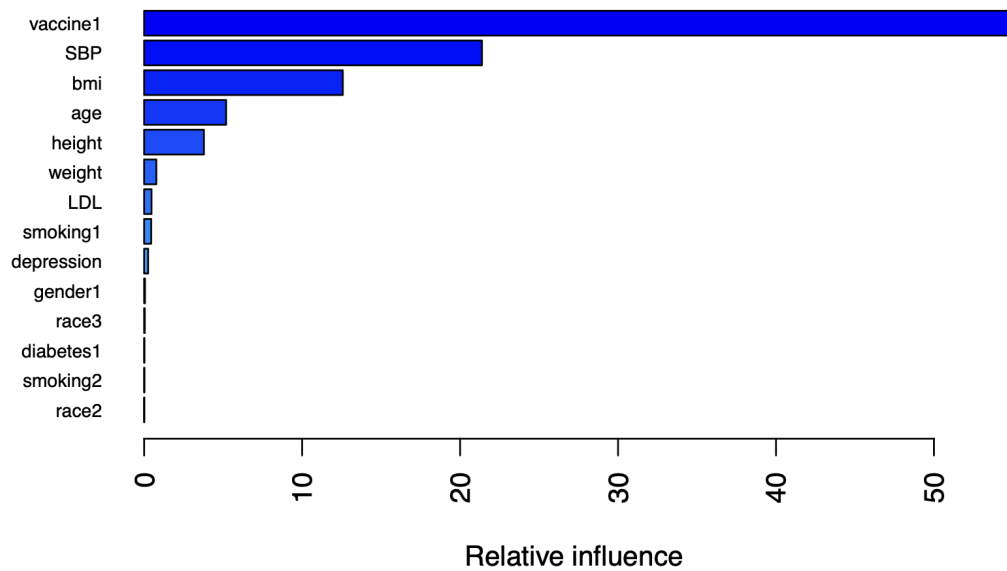


Figure 6. Partial dependence plots for the top three predictors from the AdaBoost model: (a) vaccination status, (b) systolic blood pressure (SBP), and (c) body mass index (BMI).

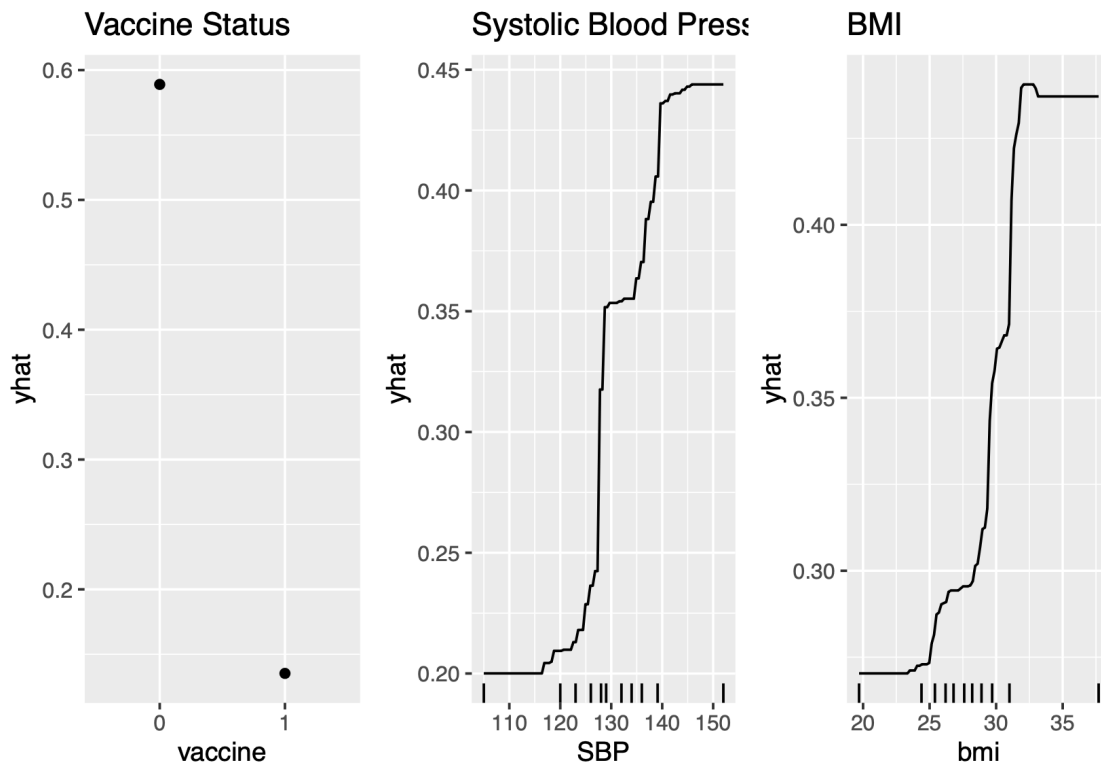


Table 3. Top 10 patients with the highest predicted probability of severe COVID-19 infection from the AdaBoost model.

Predicted Probability	True Severity
0.904	severe
0.892	severe
0.885	severe
0.874	severe
0.867	severe
0.857	severe
0.856	severe
0.820	severe
0.819	severe
0.813	severe

Table 4. Summary of predicted risk scores stratified by true severity group, based on the AdaBoost model.

Severity Group	Mean	Median	Min	Max
severe	0.602	0.700	0.085	0.904
not_severe	0.184	0.141	0.061	0.692